2 A bar is fixed between two stands. A rod of mass M is suspended from the bar by two parallel strings as shown in Figure 2.1.

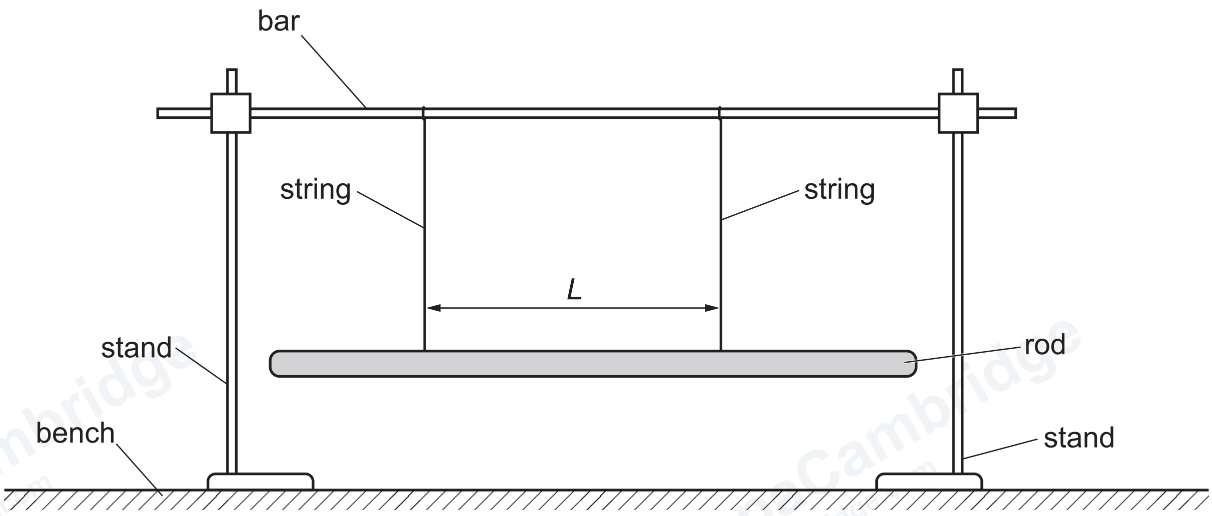


Figure 2.1

The separation of the two strings is L.

One end of the rod is displaced horizontally and then released. The rod rotates with oscillations about its centre. The time for 10 oscillations is t. The period T of the oscillations is determined.

The experiment is repeated for different values of L.

It is suggested that T and L are related by the equation

$$T = \frac{2\pi}{L}\sqrt{\frac{\alpha D}{Mg}}$$

where g is the acceleration of free fall, D is the distance between the bar and the rod, and α is a constant.

(a) A graph is plotted of T^2 on the y-axis against $\frac{1}{L^2}$ on the x-axis.

Determine an expression for the gradient.

gradient = ... [1]

(b) Values of L and t are given in Table 2.1.

Table 2.1

L / cm	t / s	$\frac{1}{L^2}$ / m^{-2}	T / s	T^2 / s^2
27.4	15.8 ± 0.2			
30.4	14.5 ± 0.2			
34.8	12.6 ± 0.2			
40.8	10.8 ± 0.2			
48.4	9.2 ± 0.2			
55.2	8.0 ± 0.2			

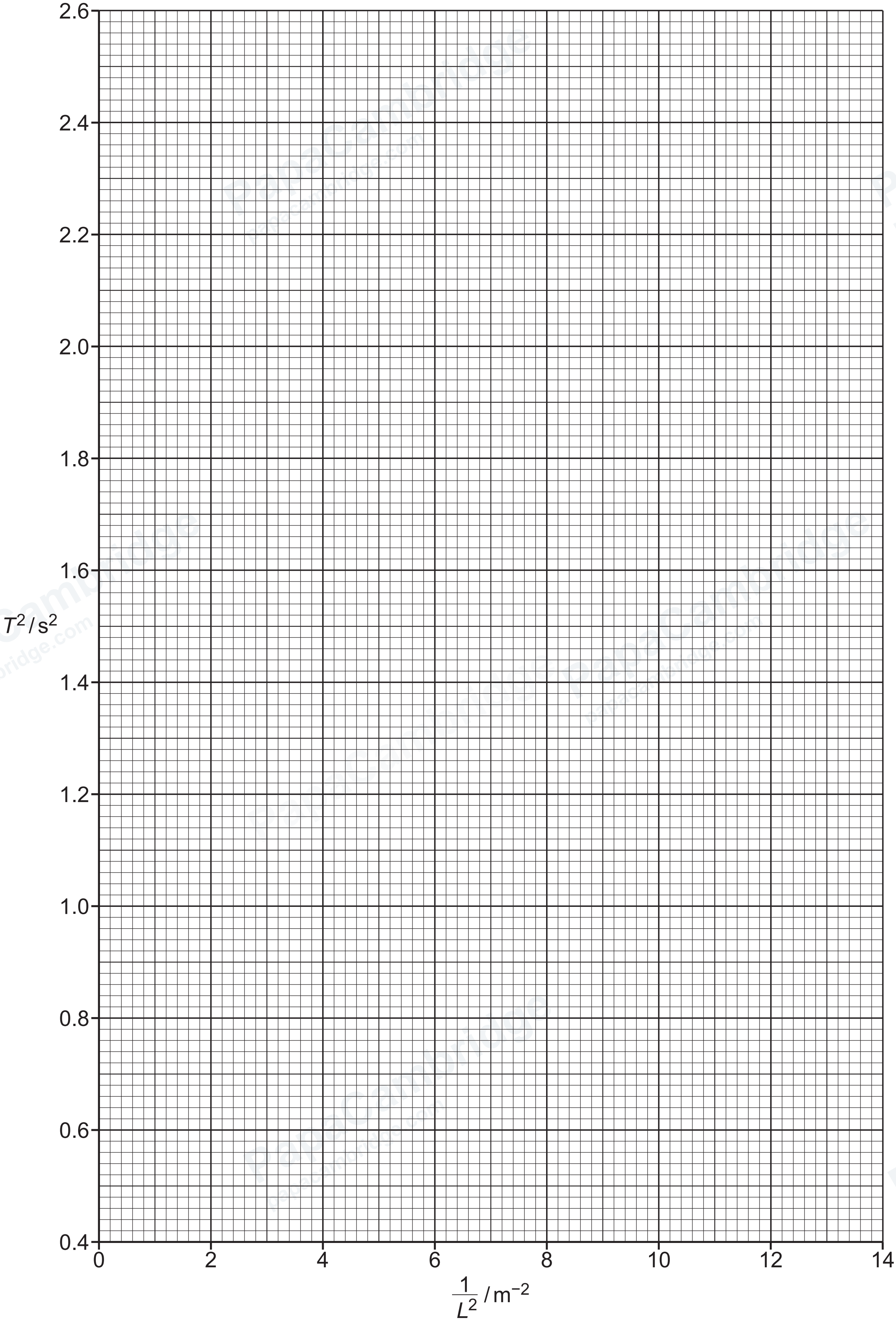
Calculate and record values of $\frac{1}{L^2}$ / m^{-2}, T / s and T^2 / s^2 in Table 2.1. Include the absolute uncertainties in T and T^2. [2]

(c) **(i)** Plot a graph of T^2 / s^2 against $\frac{1}{L^2}$ / m^{-2}. Include error bars for T^2. [2]

(ii) Draw the straight line of best fit and a worst acceptable straight line on your graph. Label both lines. [2]

(iii) Determine the gradient of the line of best fit. Include the absolute uncertainty in your answer.

gradient = ... [2]



2.6
2.4
2.2
2.0
1.8
1.6
T^2 / s^2
1.4
1.2
1.0
0.8
0.6
0.4
0
2
4
6
8
10
12
14
$\frac{1}{L^2}$ / m^{-2}

(d) The distance between the bar and the rod is measured several times:

0.826 m 0.829 m 0.824 m 0.821 m.

Determine the mean distance D. Include the absolute uncertainty.

D = ... m [1]

(e) **(i)** Using your answers to **2(a)**, **2(c)(iii)** and **2(d)**, determine the value of α. Include an appropriate unit.

Data: $g = 9.81\,\mathrm{m\,s^{-2}}$
$M = (97 \pm 2)\,\mathrm{g}$

α = ... [2]

(ii) Determine the percentage uncertainty in α.

percentage uncertainty in α = ...% [1]

(f) The experiment is repeated. Determine the separation L of the two strings that gives a value of T of $(2.20 \pm 0.02)\,\mathrm{s}$. Include the absolute uncertainty.

L = ... m [2]

[Total: 15]